Aging Mitigation in Systolic Array Accelerators: Balancing PE Loads for Enhanced Reliability

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***Abstract*—The systolic-array-based AI accelerator has been considered as one of the most promising architectures for data-intensive computation. However, the fixed systolic array size can lead to notable workload imbalance, with PEs utilization rate varying greatly, potentially reaching up to 14x disparity between maximum and minimum usages. The aging effects such as Hot Carriers Injection (HCI) and Negative Bias Temperature Instability (NBTI) can lead to timing errors or functional failures as the circuit operates for a certain while, especially in heavily utilized PEs. Consequently, the errors PE produced would propagate throughout the CNN computation, rendering the systolic array unusable and shortening its lifetime. Our paper proposes an aging-aware task deployment framework to increase utilization and mitigate aging effects. Experimental results demonstrate that our framework can outperform a conventional systolic-array-based AI accelerator 74.75% on prediction accuracy after 10 years of aging. In the meantime, our method can improve utilization up to 2.4x.**

Keywords—Systolic array, NBTI, HCI, AI accelerator, fission, data mapping

I. Introduction

Convolution Neural Network (CNN) have attracted huge attention recently due to their remarkable success in various fields, revolutionizing tasks such as image recognition, natural language processing, and medical diagnosis. The key techniques behind CNN is feature extractions from input data, which involves huge Multiply Accumulate (MAC) operations. To efficiently perform such large number of computations during inference stage, a custom-designed AI accelerators are in demand. Recently, the systolic-array-based AI accelerator [18] with the well-structured dataflow has been considered as one of the promising solutions for efficient CNN inference computing due to potential high parallelism. One classic systolic-array-based AI accelerator is Google Tensor Processing Unit (TPU), which contain 256 x 256 processing element (PE) and all PEs can perform computation simultaneously. As reported by [1], the performance of TPUv1 outperform traditional CPU 15-30 times.

The systolic-array-based AI accelerators leverage tightly coupled two-dimensional array structure and neighbor-to-neighbor dataflow to enable highly parallel and efficient computation. There are various dataflows for mapping MAC operations from a CNN model to a systolic-array-based AI accelerator. Amount them, weight stationary which stores weight in the internal register of each PE and prorogates input data from an edge of the array has been considered as one of the most promising mapping strategies for pipelining computation and minimizing data accessing from memory to improve performance [15]. Fig. 1 (a) shows an example of a weight stationary systolic array where the input data and weights are unrolled and the pipeline computations between filters can be achieved. More details will be provided in Section II. A.

Although such an architecture as well as the data mapping strategy can significantly speed-up the computation performance, the mismatch between the number of MAC operations and the number of PEs in the systolic array may degrade the system performance. The mismatch comes from the unequal filter size between layers of a CNN model. For example, in Vgg16 [19], Conv1 consists of 3x3x3x64 filters, but Conv5 consists of 3x3x128x256 filters. Assume we have 256x256 PEs in a systolic array. Obviously, the number of operations of Conv1 is smaller than the number of PEs, which make some PEs idle during the pipelined operation. On the other hand, the number of operations of Conv5 is much larger than the number of PEs, which needs multiple runs to complete its operation. Moreover, during the last run of the computation, the number of MAC operations may smaller than the number of PEs, which cause idle PEs again and degrade system performance. Such an example can be demonstrated on Fig. 1(b).

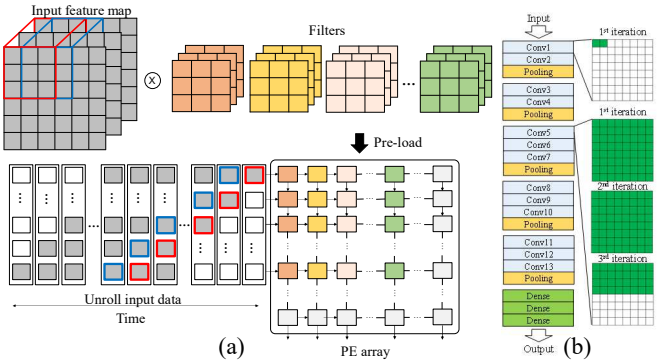


Fig. 1. (a) Dataflow of weight stationary on fixed size systolic array (b) The mismatch between sizes of the filters and the systolic array size with Vgg16 convolutions of Conv1 and Conv5.

The unbalanced PE utilization will not only cause performance degradation of a systolic array but also raise serios reliability concerns when aging effects appear. As CMOS technology continues to shrink into the nano-scale realm, circuit designs is encountering substantial reliability challenges stemming from the escalating aging effects. Hot Carriers Injection (HCI) and Negative Bias Temperature Instability (NBTI) are major aging effects where threshold voltage (V_{th}) of turned on transistors will increase over time and cause longer prorogation delay of logic gates or even functional failure of a circuit [3]. The aging effects can be slightly recovered when the circuit is idle and the transistors are turned off. Considering the

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fact that the unbalanced PE utilization in the systolic array mentioned above, the aging status of each PE may be different. Fig. 2(a) gives such an example that when a 256x256 PEs are used to execute Inception V4, the PEs on the left upper corner surfer most serious aging effects. Moreover, since the output data of a filter is obtained by accumulating all values generated by a column of PEs, such serious aged PEs will provide unreliable MAC results, leading to inaccuracy prediction of the CNN. To evaluate the 10-year drop in InceptionV4's ImageNet classification accuracy, we follow to simulate different aged scenarios of the multiplier [5]. Fig. 2 (b) shows that a significant accuracy drop happens when the imbalance-aged PEs exist as shown in Fig. 2(a).

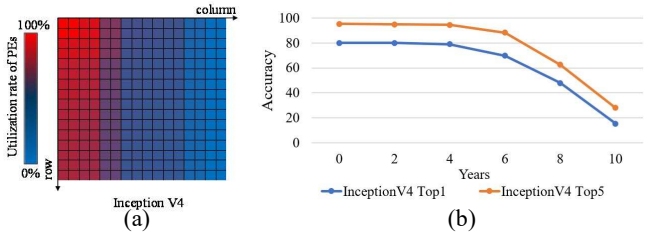


Fig. 2. (a) The utilization rate distribution of PE array with InceptionV4 (b) The accuracy of the InceptionV4 over ten years of aging.

To address the aging-induced timing failures and functional errors, previous researchers have proposed various aging mitigation and tolerance schemes [3][4][5][6][16]. Although these schemes may partially recover aging-induced accuracy drops of CNN models, without considering the unbalanced PE utilization still makes the room for further improvement. We find that the accuracy drops of CNN models can be minimized if the PEs are equally aged and the mapping of MAC operations to PEs are tactically performed.

In this paper, we introduce a pioneering aging-aware task deployment framework tailored for systolic-array-based AI accelerators. Our framework comprises two primary components: a balance-loaded systolic array datapath design and an aging-aware data mapping policy. The balance-loaded systolic array datapath facilitates the efficient utilization of idle Processing Elements (PEs) within the array, enhancing PE utilization rates while accelerating computations. Leveraging the concept of fission structure [20], input data can be flexibly fed into the systolic array, effectively utilizing idle PEs. Meanwhile, the aging-aware data mapping policy allocates workloads to slightly aged PEs to ensure symmetric aging. Ensuring that each PE is utilized evenly minimizes the risk of serious aging, enhancing system reliability. Here we summarize our contributions in this paper as follows:

1. To address accuracy degradation in systolic-array-based AI accelerators due to aging, we introduce a novel aging-aware task deployment framework. This framework includes a balance-loaded systolic array datapath design and an aging-aware data mapping policy, aimed at preventing uneven PE aging.
2. In our balance-loaded systolic array datapath design, we implement a fission structure, adding peripheral circuits within defined area constraints to maximize PE utilization rates. This design enhances infrastructure for utilizing the majority of PEs, effectively distributing aging effects across all PEs.
3. In our aging-aware data mapping policy, we propose a heuristic-based algorithm that considers aging status of each PE, and perform skilled data-to-PE mapping for aging balance.
4. Experimental results show that after 10-year aging, our framework can outperform a conventional systolic-array-based AI accelerator 74.75% on prediction accuracy.

II. Background

In this section, we firstly provide the basic concepts of systolic-array-based AI accelerators, following by introducing how the CNN inference can be performed on the weight stationary dataflow. After that, we study the aging-induced threats on systolic-array-based AI accelerators, and review some existing articles for aging mitigation and tolerance.

A. Systolic-array-based AI Accelerator

Systolic-array-based AI accelerator is specialized processor designed to efficiently perform general matrix multiplication (GEMM) [9], particularly in applications such as deep neural network processing. As shown in Fig. 3 (a), Systolic-array-based AI accelerator typically consist of a two-dimensional array of PE, weight buffer, global buffer, and accumulation unit. Each PE has internal storage and computational components dealing with MAC operations. Despite the independent computation of each output value, the peer-to-peer links enable efficient data reuse and parallel computation. The Google TPU [1], a well-known example of a systolic array accelerator used for weight-stationary matrix operations.

B. Weight Stationary

In weight stationary dataflows [10], each PE stores weights, minimizing weight movement within the array. This strategy effectively reduces data access overhead from memory while spatially broadcasting activations and accumulating partial sums across the PE array. The process involves preloading weights into the array before processing input data. As shown in Fig. 3 (b), input data is pipelined through the array, with each PE performing computations and passing results to the next PE. This pipeline structure facilitates continuous processing, enabling each PE to start computation immediately upon receiving data from the previous PE, without waiting for the entire matrix operation. Our method builds upon weight stationary principles, offering flexibility for various neural network topologies by separating weights from activations, thus accommodating diverse layer types and network configurations.

In cases where the filter size surpasses the capacity of the systolic array, the filter needs to be partitioned into smaller segments, referred to *sub-mappings* in this paper. This partitioning allows the systolic array to accommodate larger filters by performing multiple iterations. The size of the filter is influenced by three main factors: the kernel size, input channel size, and output channel size. The horizontal dimension is determined by the output channel size, while the vertical dimension is calculated as the product of the input channel size and the kernel size. In summary, mismatches may occur more frequently in the final iteration due to the need to maximize PE usage. By carefully managing sub-mappings and considering the interplay of kernel size, input and output channel sizes, and PE

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utilization, the systolic array can effectively handle filters that exceed its capacity.

C. Aging Effect on DNN Accelerator

NBTI and HCI are fundamental aging phenomena affecting transistors. In both NBTI and HCI phenomena, turning transistor on would lead to V_{th} increase and the current in the ON state of the transistor (Ion) reduction. While the circuit keeps operating,

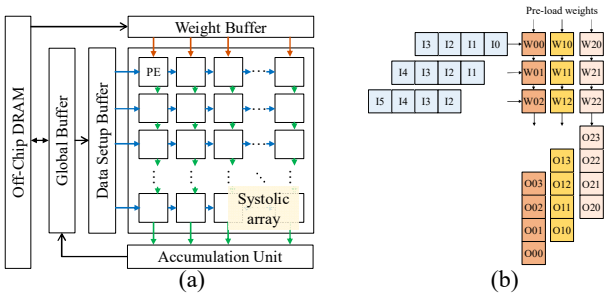


Fig. 3. (a)Weight stationary systolic-array-based AI accelerator (b) Pipeline structure of systolic array.

the aging effects accumulate in the transistor, resulting in a timing error. Shrinking transistor sizes exacerbate NBTI and HCI effects, recent evaluation indicates approximately a 10% delay increase after 3 years [11]. Moreover, after 7 years, two distinct aging rates, labeled as Low Wearout and High Wearout, demonstrate respective 10% and 25% increases in delay [12]. The conventional approach involves incorporating the timing guardband during the design phase, but this tends to reduce the frequency. While reducing the frequency can address the issue, during the initial stages, the circuit doesn't necessitate this additional time, resulting in unnecessary waste.

This work adopts a predictive model for NBTI and HCI aging effects, relying on a significant prior study[12][13]. The model estimates the NBTI-induced $V_{th,NBTI}$ and HCI-induced $V_{th,HCI}$, considering model parameter (E_a and E_b), technology-specific factor ($A_{NBTI}$and A_{HCI}), operating voltage (Vdd), time (t), frequency (f), and duty cycle (d, which is equivalent to the utilization rate u of an PE).

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{\left(\frac{-E_a}{kT}\right)} \times V_{dd}^4 \times t^{1/6} \times u^{1/6} \quad (1)$$

$$\Delta V_{th,HCI} = A_{HCI} \times u \times f \times e^{\left(\frac{-E_b}{kT}\right)} \times t^{1/6} \quad (2)$$

Clearly, except for model parameter, the utilization rate of PE and temperature have a significant impact on NBTI and HCI effects. This points to that optimizing the mapping of convolutions onto the systolic array could effectively mitigate the aging issue on the accelerator., which means that we can optimize mapping of convolutions onto the systolic array could effectively mitigate the aging issue on the accelerator.

D. Related Works

The related works contribute to the advancement of systolic-array-based AI accelerators by addressing fault tolerance, timing error, and design improvements in deep learning hardware accelerators. To maintain the functionality while systolic-array-based AI accelerator aged, [4][16] are proposed to deal with the permanent faults in the systolic array. SalvageDNN [4] propsed a technique that salvages faulty deep neural network accelerators by using saliency-driven fault-aware mapping to enable reliable execution despite hardware faults. To mitigate the timing error of AI accelerator, [5][6] are proposed to deal with it. Salamin et al [5] achieves slightly degrades inference accuracy over time while compensating for the aging-induced delay increase in the NPU by quantizing the input data of neural network model. While past endeavors have effectively improved the functionality of systolic arrays post-aging, prior research has overlooked the importance of optimizing the distribution of PE utilization rates to efficiently combat the impacts of aging.

III. Problem Formulation

In this section, we firstly list all of the assumptions made within this paper, and formulate the problem we address. First, we target the systolic-array-based AI accelerator with weight stationary dataflow. Secondly, we concentrates on the reliability threats posed by NBTI and HCI.

Here we formally formulate the aging-aware task deployment framework tailored for systolic-array-based AI accelerators as follows: Given a systolic-array-based AI accelerator with N x N PEs and k CNN models where the model M_x consists of L_{M_x} layers and the yth layer contains z filters. We want to find a MACs-to-PEs mapping methodologies such that the utilization rate of each PE can be balanced and the aging-induced accuracy drop of CNNs can be minimized.

IV. Framework

A. Overview

In this paper, an aging-aware task deployment framework for the systolic-array-based AI accelerator is proposed. Again, the proposed framework aims to balance the utilization rate of each PE so that the aging-induced accuracy drop of CNN can be minimized. The major concept behind this framework is to parallelly execute MAC operations that involves different input data. As shown in Fig. 3(a), input data is sent from the left edge of the array and prorogate all the way to the right edge. Therefore, only filters that applies to the same input data can be performed, which leads to potential idle PEs at right portion of the array. If we can slit the array into multiple fissions and each fission can accept input data independently, the filters with different input data can be performed simultaneously and the utilization rate of each PE can be balanced. However, to realize such an idea, we have two challenges. First, extra hardware cost is needed to send input data to each fissions, which may lead to extra design cost. Second, a sophisticated MACs-to-PEs strategy is in demand to balance PE utilization when the circuit is on-the-fly. Therefore, our framework is a two-stage approach which consists of (1) a balance-loaded systolic array datapath design and (2) an aging-aware data mapping policy, as shown in Fig. 4. We will detail each step in the following sub-sections.

B. Fission systolic array hardware design

The fission systolic array structure can utilize the idle PEs to enhance the utilization and balance the computing loads across the entire systolic array. The fission systolic array structure can support parallelly computation of multiple sub-mappings together in the systolic array by importing corresponding input data and filters as the Fig. 5 shown. Note that horizontal distribution of computation load is resolved through the balance-

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loaded systolic array datapath design, while the vertical distribution is addressed via the aging-aware data mapping policy.

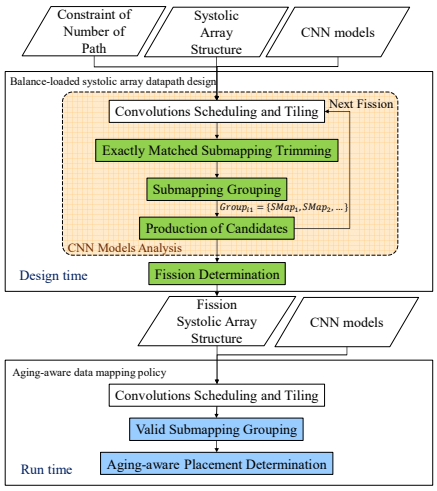


Fig. 4. Overview of the framework.

In the PE design, PEs at the fission need to add the mux to choose whether the input should come from its left or the buffer. By doing so, for large sub-mappings, the input port can choose left to share the same input data for the same row. By changing the input source, the hardware can flexibly configure for different sizes of the sub-mappings. For example, connect Path 2 and Path 3 to left can utilize the whole array for a large sub-mapping. Connecting Path 2 to left and Path 3 to the input buffer can realize one big sub-mappings and one small sub-mappings start from Path 3. Furthermore, by setting the zero weight of the idle PEs in the same column during weight preloading process, the PEs skip its computation (i.e. at idel mode) and transit the partial sum to the next PE, which mitigate aging effects on zero-weighted PEs. When there is sparsity in the weight, PE can save power consumption [17]. Finally, the balance-loaded systolic array datapath structure extends the Data Setup buffer unit of the baseline TPU architecture to feed input data for each path.

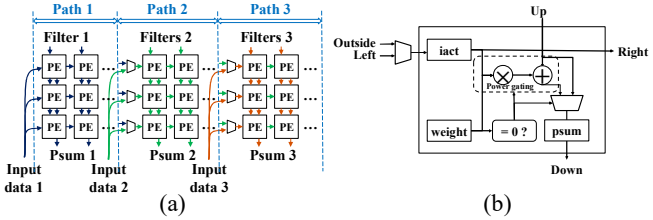


Fig. 5. (a) The fission dataflow systolic array structure (b) The design of PE at the fission position.

C. Balance-loaded systolic array datapath design

Proper positions of fission achieve higher utilization and provide better solution space for mapping convolutions onto the systolic array. However, because the sizes of CNN models exhibit variability, from kernel size, input channel size, and number of filters, the proper positions of fission differ from one layer to another. Consequently, as illustrated in the Fig. 4, the system analyzes all of the CNN models to identify the most suitable positions of fission. Finally, the system determines the position of the fission depends on the result of the analyze.

IV.C.1. CNN models analysis

To fully utilize the systolic array while considering all of the CNN models, the system imitates the aging-aware data mapping policy. The system arranges the sub-mappings in the same time slots, explores all of the combination, and calculates the highest estimated utilization (EU) [14]. Because the position of fission varies with the order of sub-mappings and differs between groups of sub-mappings, permutation of all sub-mappings in the group is necessary to generate the mapping candidates. In Fig. 6, consider two convolutions scheduled in the same timeslots $T_i = \{conv_{i1}, conv_{i2}\}$ as an example. For $conv_{i1}$, to computing a value of output feature map, 3x3x80 input data need to multiply with the 3x3x80 filter and accumulate into one number. For 256x256 systolic array, it needs 3 sub-mappings which denoted as $\{SMap_A, SMap_B, SMap_C\}$ to complete. For $conv_{i2}$, its kernel size, input channel, number of filters are 5x5, 16, and 48, respectively. Hence, for 256x256 systolic array, it needs 2 sub-mappings which denoted as $\{SMap_D, SMap_E\}$ to complete.

IV.C.2. Sub-mappings grouping

In grouping Map, we utilize the dynamic programming to minimize the computation. The system sort Map from highest of number of filters to lowest number of filters $\{SMap_A, SMap_B, SMap_C, SMap_D, SMap_E\}$. Then, the first Map will group with the second Map. If the sum of the number of filters is smaller than the systolic array size, the combination will get recorded and add the next Map. If the sum of the number of filters is greater than the systolic array size, the system will skip to the next combination. For example, the sum of number of filters of $\{SMap_A, SMap_B\}$ or $\{SMap_A, SMap_C\}$ are greater than array size. However, the sum of number of filters of $\{SMap_A, SMap_D\}$ is equal to the systolic array size. Then, the system will calculate the EU [14] of this combination, which is 79.03%, and rrecord it as a potential solution. The system also considers the difference between different sub-mappings' column sizes, as this can affect the input data loading time. It evaluates this by calculating the EU, and chooses the combination with the highest EU as the final solution. After find all of the valid group, the system will choose from the highest estimated utilization of the combination.

$$Estimated\ Utilization = \frac{\sum Num.\ activated\ PEs}{Num.total\ PEs \times cycles} \quad (3)$$

$$cycles = cycles_{calculating} + cycles_{empty} \quad (4)$$

	IFM height	IFM width	Input channel	Kernel x	Kernel y	Output channel
Conv 1	28	28	80	3	3	208
Conv 2	28	28	16	5	5	48

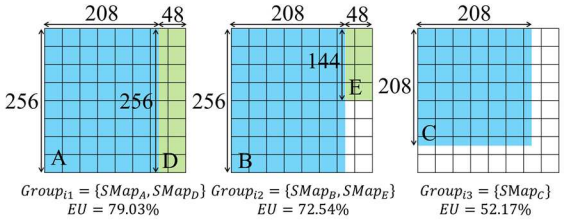


Fig. 6. Example of grouping mappings of convolutions.

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IV.C.3. Production of Candidates

The system permutes the sub-mappings in the same group to produce the candidates. For example, in Fig. 7 shown below, the grouped sub-mappings can produce the candidates $(SMap_A, SMap_D)$ and $(SMap_D, SMap_A)$. The corresponding possible position of fissions would be position 49 and 209. Subsequently, the system recorded all of the possible positions of fissions and selects the most compatible position of fission for the systolic array based highest selected position. The system continues this process and selects positions for the fissions one by one until the system reach the constraints of the number of the entries.

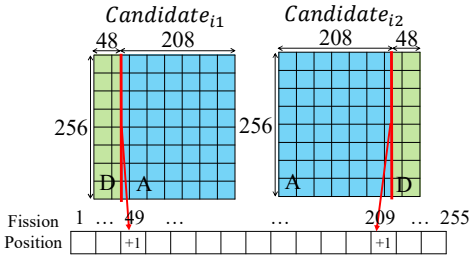


Fig. 7. Example of generating candidate mapping and record the fission positions.

D. Aging-aware data mapping policy

To further ensure the balance of the systolic array, we will use the aging-aware data mapping policy to balance the computing loads of PEs in the systolic array. As illustrate in Fig. 4, producing the candidate is the same as the balance-loaded systolic array datapath design. The difference is that with the fixed position of fisssions, the system will filter the invalid candidates. For example, it is impossible to map the candidate $(SMap_D, SMap_A)$ onto fisssion systolic array with position of fission at {128, 209}. After producing all the valid candidates, we can place our sub-mappings while considering the computation load distribution. For example, in Fig. 8, there are the chosen candidate $(SMap_G, SMap_F)$ and the stress table which recorded usage of the systolic array. Since the convolution size of CNN models usually would be the multiple of 16, 32, or 64, the computing loads nearby will be the same. Hence, to reduce the complexity of the stress table, the system will evaluate the CNN models to reduce the size of workloads table, for example reduce the size from 256x256 to 8x8. In determining sub-mappings placement, the system places big number of the activated PEs sub-mappings first. Then, the system would enumerate the start rows and columns of the sub-mappings and calculate the sum of the stress with PEs are used for the sub-mappings. After placing sub-mappings, the system would eliminate the invalid start of the rest of sub-mappings. The system continues this process until all of the sub-mappings are placed.

When the candidate has only one sub-mappings, the system utilizes the fission systolic array structure to accelerate the computation. For small sub-mappings, the system copies the filter to available PEs and split the input data to parallelly compute the convolution. For large sub-mappings, the system utilizes the fission to import the input data to avoid the loading time. In (3), we can know that reduce the cycles of preload can enhance the utilization.

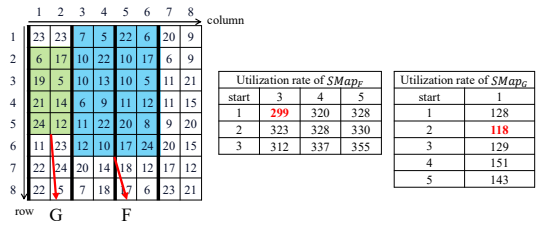


Fig. 8. Example of placing Map onto the systolic array.

V. Experimental Results

To evaluate the effectiveness of aging-aware task deployment framework in balancing the utilization rate of PE and maximizing the utilization of systolic array, we conduct a series of experiments. We implement the balance-loaded systolic array datapath design and aging-aware data mapping policy with C++ to find the proper balance-loaded systolic array datapath configuration and the optimal position of sub-mappings. We use Google's TPU[1] as the base hardware for accelerating DNNs. For aging analysis, similar to prior work [5], we focus on NBTI and HCI aging of the MAC unit, a critical circuit driving performance. We leverage the aging model described in Section II (Eq. 1, 2) and examine a typical 10-year lifespan. Moreover, the accuracy loss is calculated with respect to the accuracy achieved with FP32 inference.

A. Balance-loaded systolic array datapath validation

To verify the effectiveness of balance-loaded systolic array datapath design, we calculate the probability of utilization rate of PEs by placing every sub-mappings using all of the possible fission. We further demonstrate the impact by showing the variance in PE utilization rate probability across different models. High variance indicates the structure of systolic array that cannot be compatible with every sub-mappings, result in low overall utilization. Additionally, it suggests an unbalanced distribution of workload across PEs without the aging-aware data mapping policy. The Fig. 9 illustrates that the variance in PE utilization rate probability within the conventional systolic array is 3.3x higher compared to the four-fission systolic array in Dpn68. This variance steadily decreases with the addition of new fissions. The increased number of fissions provides more flexibility in utilizing the entire array for accommodating convolution mappings of various sizes.

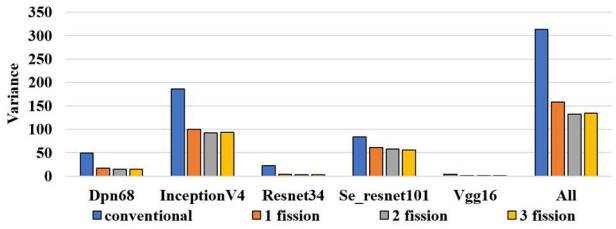


Fig. 9. The probability of utilization rate of PEs.

B. Aging-aware data mapping policy validation

Fig. 10 demonstrates that after applying the aging-aware data mapping policy, convolution operations can be precisely mapped onto chosen positions within the fission systolic array structure. In the Fig. 10, as the number of fissions increases,

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utilization also rises, maximizing the use of the entire systolic array. The result aligns with prior research [14] suggesting that minimizing input data loading time improves utilization. The introduction of fission creates additional shortcuts for loading the input feature maps, leading to minimized idle PEs and ultimately improving overall utilization. Our findings reveal that fissions effectively enhance utilization across various CNN architectures, including those with branch structures, like InceptionV4 and Dpn68, and those without, like Vgg16 and Resnet34. Fig. 11 shows that the utilization rate of PEs in the conventional systolic array has a higher variance compared to the fission systolic array. A higher variance in utilization rate indicates a greater imbalance in the workload distribution among PEs. Therefore, our proposed method achieves a 3.6x improvement in variance of utilization rate compared to the conventional systolic array, considering all CNN models. Additionally, the range of PE utilization rate in our work shows a 3.7x improvement over the conventional approach across all CNN models.

Due to space limitations, we present the accuracy drop of InceptionV4, during the ten years of aging, the accuracy will decrease based on the aging model. InceptionV4 within the conventional systolic array experiences a significant accuracy drop, with the drop of up to 81% in InceptionV4. Our method still outperforms the conventional systolic array by 74.75% in InceptionV4 after ten years of aging.

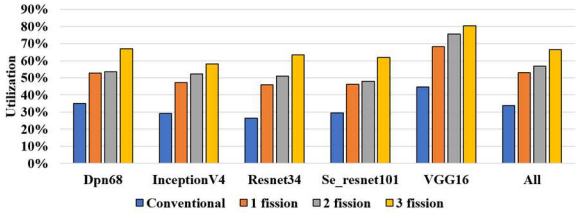


Fig. 10. Utilization of systolic array utilizing.

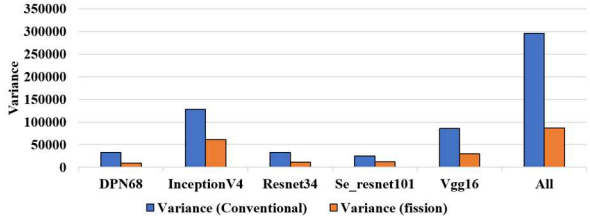


Fig. 11. Utilization rate of PEs.

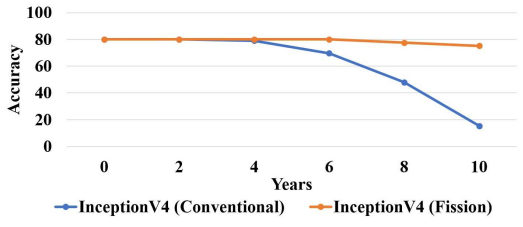


Fig. 12. Inference Accuracy over ten years of aging.

C. *Overhead analysis*

Here we estimate the area overhead of our framework under different configurations of fission with transistor count. In the Fig. 13, we can find the area overhead will increase with the increasing of the number of the fission. However, as previous section mentioned, the variance of probability of utilization rate of PEs would drop to some level and saturate as the number of fission increase, while the utilization would increase.

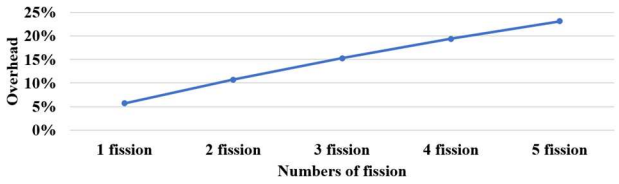


Fig. 13. Area over head of our framework.

VI. Conclusions

This work presented an aging-mitigate design framework for systolic-array-based AI accelerators. It tackles workload imbalance and aging effects through balance-loaded systolic array datapath design and aging-aware data mapping policy enhancement. Simulations show our framework can outperform a conventional systolic-array-based AI accelerator 74.75% on prediction accuracy after 10 years of aging. The framework achieves a 2.4x performance improvement due to fission functionality. These results demonstrate the promise of our framework for extending accelerator lifetime and enhancing DNN performance.

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